

Week 4: Exercises

Complete all 10 exercises. **For each one, write pseudocode first, then write the JavaScript.**

Exercise 1: FizzBuzz

Print numbers 1-100. But: if the number is divisible by 3, print “Fizz”. If divisible by 5, print “Buzz”. If divisible by both, print “FizzBuzz”.

Pseudocode hint:

```
FOR i FROM 1 TO 100
  IF i divisible by 3 AND 5 THEN
    PRINT "FizzBuzz"
  ELSE IF i divisible by 3 THEN
    PRINT "Fizz"
  ELSE IF i divisible by 5 THEN
    PRINT "Buzz"
  ELSE
    PRINT i
  ENDIF
ENDFOR
```

Important: Check the “both” condition first. If you check “divisible by 3” first, numbers like 15 will print “Fizz” and never reach the “FizzBuzz” check.

Solution

```
```javascript
for (let i = 1; i = 100; i++) {
 if (i % 3 === 0 && i % 5 === 0) {
 console.log("FizzBuzz");
 } else if (i % 3 === 0) {
 console.log("Fizz");
 } else if (i % 5 === 0) {
 console.log("Buzz");
 } else {
 console.log(i);
 }
}
```
```

Exercise 2: Even or Odd

Given a number, print whether it's "Even" or "Odd".

Pseudocode hint:

```
SET number TO 7
IF number divided by 2 has remainder 0 THEN
PRINT "Even"
ELSE
PRINT "Odd"
ENDIF
```

Solution

```
```javascript
let number = 7;

if (number % 2 === 0) {
 console.log("Even");
} else {
 console.log("Odd");
}
```
```

Try changing the number to 4, 0, -3. Does it still work?

Exercise 3: Largest of Three Numbers

Given three numbers, find and print the largest one.

Pseudocode hint:

```
SET a TO 10, b TO 25, c TO 15
IF a > b AND a > c THEN PRINT a
ELSE IF b > a AND b > c THEN PRINT b
ELSE PRINT c
```

Solution

```
```javascript
let a = 10;
let b = 25;
```

```
let c = 15;
```

```
if (a > b && a > c) {
 console.log(a + " is the largest");
} else if (b > a && b > c) {
 console.log(b + " is the largest");
} else {
 console.log(c + " is the largest");
}
...
```

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## Exercise 4: Count Vowels

Count how many vowels (a, e, i, o, u) are in a word.

### Pseudocode hint:

```
SET word TO "education"
SET count TO 0
FOR EACH letter IN word
 IF letter is a vowel THEN
 ADD 1 TO count
 ENDIF
ENDFOR
PRINT count
```

### Solution

```
```:javascript  
let word = "education";  
let count = 0;  
  
for (let i = 0; i < word.length; i++) {  
  let letter = word[i].toLowerCase();  
  if (letter === 'a' || letter === 'e' || letter === 'i' || letter === 'o' || letter === 'u') {  
    count = count + 1;  
  }  
}  
  
console.log("Vowels: " + count); // 5  
...
```

`**.toLowerCase()` converts the letter to lowercase so "A" and "a" both match.

Exercise 5: Reverse a String

Take a string and print it backwards. "bootcamp" becomes "pmactooB".

Pseudocode hint:

```
SET word TO "bootcamp"
SET reversed TO ""
FOR i FROM last index DOWN TO 0
  ADD character at i TO reversed
ENDFOR
PRINT reversed
```

Solution

```
````javascript
let word = "bootcamp";
let reversed = "";

for (let i = word.length - 1; i >= 0; i--) {
 reversed = reversed + word[i];
}

console.log(reversed); // "pmactooB"
````
```

****Key idea:**** Start the loop at the end (`word.length - 1`) and go backwards (`i--`).

Exercise 6: Star Triangle

Print a right-angled triangle of stars, 5 rows tall:

```
*
**
***
****
*****
```

Pseudocode hint:

```
FOR row FROM 1 TO 5
SET line TO ""
FOR star FROM 1 TO row
ADD "*" TO line
ENDFOR
PRINT line
ENDFOR
```

Solution

```
````javascript
let rows = 5;

for (let row = 1; row = rows; row++) {
 let line = "";
 for (let star = 1; star = row; star++) {
 line = line + "*";
 }
 console.log(line);
}
````
```

This uses a **nested loop** — a loop inside a loop. The outer loop controls rows, the inner loop controls how many stars go on each row.

Exercise 7: Sum 1 to N

Calculate the sum of all numbers from 1 to N (where N = 10).

Pseudocode hint:

```
SET n TO 10
SET total TO 0
FOR i FROM 1 TO n
ADD i TO total
ENDFOR
PRINT total
```

Solution

```
````javascript
let n = 10;
let total = 0;
```

```
for (let i = 1; i = n; i++) {
 total = total + i;
}

console.log("Sum: " + total); // 55
...
```

This is the **accumulator pattern**: start `total` at 0, add to it each loop cycle.`

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## Exercise 8: Palindrome Check

Check if a word reads the same forwards and backwards. "racecar" is a palindrome. "bootcamp" is not.

### Pseudocode hint:

```
SET word TO "racecar"
SET reversed TO ""
FOR i FROM last index DOWN TO 0
 ADD character at i TO reversed
ENDFOR
IF word EQUALS reversed THEN
 PRINT "Palindrome"
ELSE
 PRINT "Not a palindrome"
ENDIF
```

### Solution

```
````javascript  
let word = "racecar";  
let reversed = "";  
  
for (let i = word.length - 1; i >= 0; i--) {  
  reversed = reversed + word[i];  
}  
  
if (word === reversed) {  
  console.log("Palindrome!");  
} else {  
  console.log("Not a palindrome");  
}  
...
```

****Reuse!**** This builds on the string reversal from Exercise 5. That's pattern recognition in action.

Exercise 9: Factorial

Calculate the factorial of a number. $5! = 5 \times 4 \times 3 \times 2 \times 1 = 120$.

Pseudocode hint:

```
SET n TO 5
SET result TO 1
FOR i FROM 1 TO n
  MULTIPLY result BY i
ENDFOR
PRINT result
```

Important: Start `result` at 1, not 0. Anything multiplied by 0 is 0.

Solution

```
``javascript
let n = 5;
let result = 1;

for (let i = 1; i = n; i++) {
  result = result * i;
}

console.log("Factorial: " + result); // 120
...`
```

Exercise 10: Number Guessing Game

A secret number is 7. The “player” gets 3 guesses (simulated with an array). If they guess right, print “You win!” and stop. If all 3 guesses are wrong, print “You lose!”

Pseudocode hint:

```
SET secret TO 7
SET guesses TO [3, 7, 10]
SET won TO FALSE
FOR EACH guess IN guesses
  IF guess EQUALS secret THEN
    PRINT "You win!"
    SET won TO TRUE
  STOP LOOP
ELSE
  PRINT "Wrong guess"
ENDIF
ENDFOR
IF won is FALSE THEN
  PRINT "You lose!"
ENDIF
```

Solution

```
````javascript
let secret = 7;
let guesses = [3, 7, 10];
let won = false;

for (let i = 0; i < guesses.length; i++) {
 if (guesses[i] === secret) {
 console.log("You win!");
 won = true;
 break;
 } else {
 console.log("Wrong guess: " + guesses[i]);
 }
}

if (won === false) {
 console.log("You lose! The secret was " + secret);
}
````
```

`break` stops the loop immediately. No point checking more guesses after you've won.

Try changing the guesses array to `[3, 10, 5]` — you should see "You lose!"